Common Probability Distributions

Learning Objectives

By the end of this session, you will be able to:

- Understand the concept of random variable and probability distributions.
- Understand Normal, Binomial, and Poisson Distributions.
- Learn the formulas, attempt quizzes and solve numerical problems.
- How to work with probability distributions using Python!

Random Variable Example

- Number of sixes that will be hit in the last over of a T20 match?

Random Variable Example

- Number of sixes that will be hit in the last over of a T20 match?

No of sixes	0	1	2	3	4	5	6

- A random variable is a variable that takes on different values based on the outcome of a random event.

- It can be thought of as a function that assigns numerical values to each possible outcome of a random process.

Types of Random Variable

- **Discrete Random Variable:** Takes on a countable number of distinct values (e.g., number of heads in a series of coin flips).

- **Continuous Random Variable:** Takes on an infinite number of possible values within a given range (e.g., the exact height of students in a class).

Types of Distributions

Discrete Distributions: Binomial, Poisson

Continuous Distributions: Normal, Exponential

What is a Probability Distribution?

- A probability distribution is a function that describes the likelihood of outcomes in a sample space.

- Examples include: Normal, Binomial, Poisson

What is a Probability Distribution?

Ex: Consider the random experiment of tossing two coins.

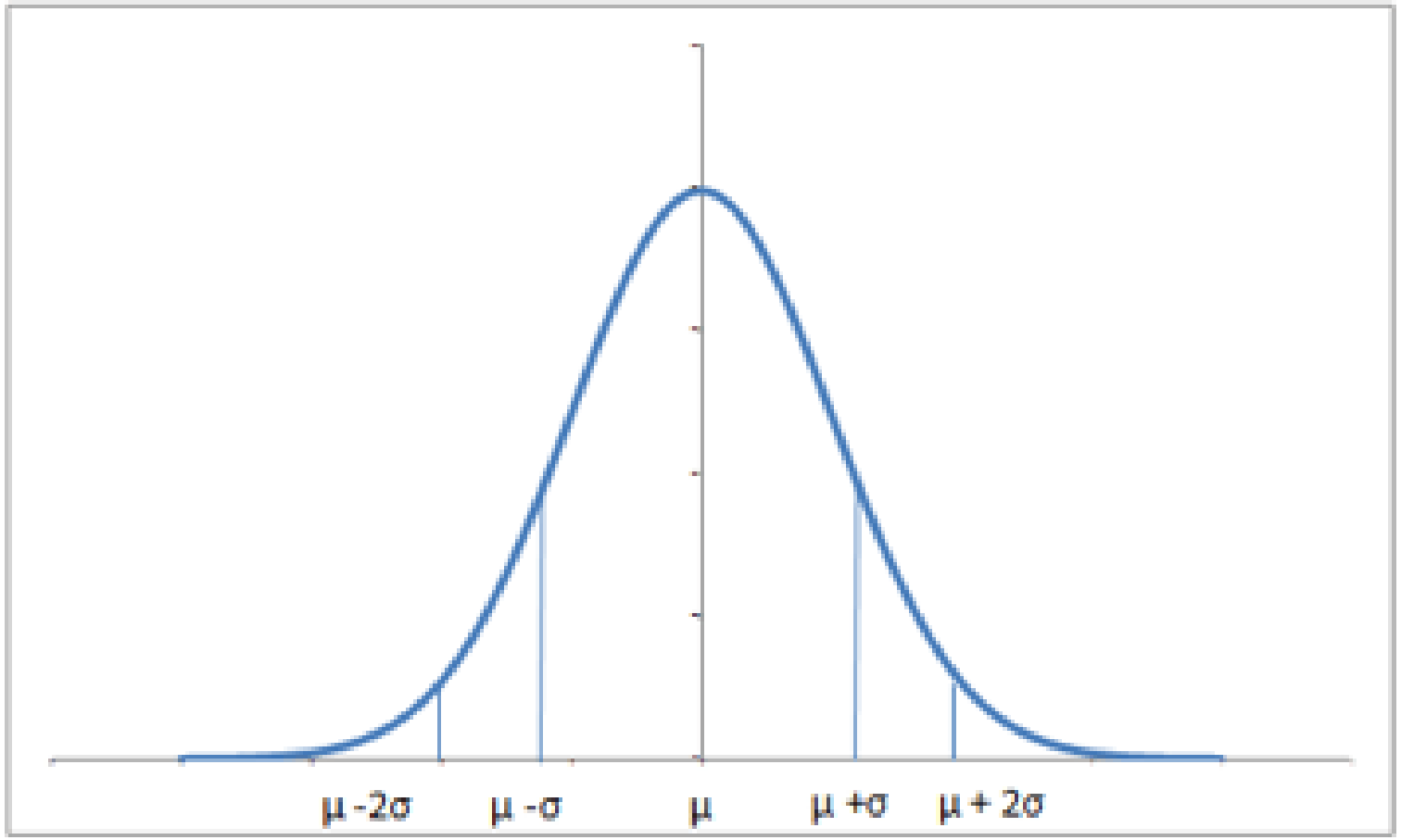
Let the occurrence of heads to be counted. There may be 0, 1 or 2 heads.

Possible outcomes	TT	HT	TH	HH
Number of heads	0	1	1	2
Probability	¼	¼	¼	¼

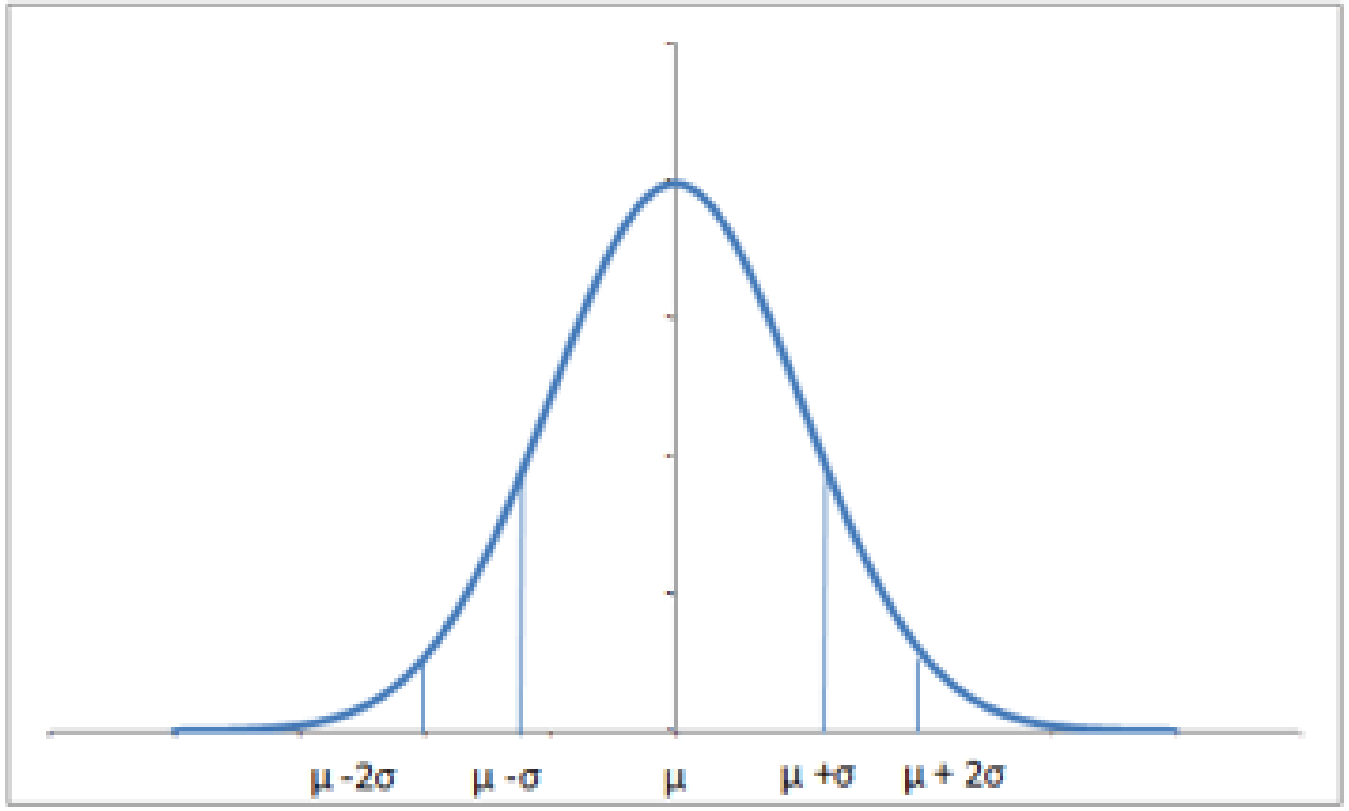
A probability distribution describes how the values of a random variable are distributed. It provides a complete description of the likelihood of different outcomes.

Normal Distribution

- A continuous distribution with a symmetric, bell-shaped curve.
- Defined by mean (μ) and standard deviation (σ).



Normal Distribution - formula



$$f(x|\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- Mean = Median = Mode
- 68-95-99.7 Rule
- Symmetrical shape

Normal Distribution - Example

- If height ~ N(170, 10), what is the probability a person is between 160 and 180 cm?

Normal Distribution – Example Solution

- If height ~ N(170, 10), what is the probability a person is between 160 and 180 cm?

- $Z = (x - \mu)/\sigma \rightarrow Z = (180 - 170)/10 = 1$
- $P(-1 < Z < 1) = 68\%$
- Hence, 68% of people are between 160 and 180 cm

Binomial Distribution

- Models the number of successes in n independent trials with probability p.
- Used for yes/no outcomes.

The probability mass function (PMF) for a Binomial distribution is:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Where:

- $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ is the binomial coefficient, representing the number of ways to choose k successes out of n trials.
- p^k is the probability of having k successes.
- $(1-p)^{n-k}$ is the probability of having $n-k$ failures.

Binomial Distribution Properties

- Discrete distribution
- Fixed number of trials (n)
- Constant probability (p)
- Independent trials

Binomial Distribution example

In an inspection of cement companies 60% of all industries had emissions that do not meet the pollution requirement. For a random sample of 15 companies, compute the following

1. Probability that all 15 fails the inspection
2. Exactly 8 fails the inspection

Binomial Distribution example

In an inspection of cement companies 60% of all industries had emissions that do not meet the pollution requirement. For a random sample of 15 companies, compute the following

1. Probability that all 15 fails the inspection
2. Exactly 8 fails the inspection

Given:

n = 15

p = 0.60

q = 1-p = 1-0.6 = 0.4

Probability that all 15 fails the inspection

Here x = 15

$P(x) = {}^{n}C_{x}\, p^{x} q^{n-x}$, x: 0,1,2….,n

$P(x=15) = {}^{15}C_{15}\, (0.6)^{15} (0.4)^{0}$

P(X=15) = 0.00047

Binomial Distribution example

In an inspection of cement companies 60% of all industries had emissions that do not meet the pollution requirement. For a random sample of 15 companies, compute the following

1. Probability that all 15 fails the inspection
2. Exactly 8 fails the inspection

Given:

n = 15

p = 0.60

q = 1-p = 1-0.6 = 0.4

Probability that exactly 8 fails the inspection

Here x = 8

$P(x) = {}^{n}C_{x}\, p^{x} q^{n-x}$, x: 0,1,2….,n

$P(x=8) = {}^{15}C_{8}\,(0.6)^{8}(0.4)^{15-8}$

$P(x=8) = {}^{15}C_{8}\,(0.6)^{8}(0.4)^{7}$

P(X=8) = 0.17708

Poisson Distribution

Sometimes there may be a situation that out of a large number of trials, probability of getting a success may be very small i.e., the event may be a rare event.

Ex: Cyber attack, Fraud, Probability of a tree falling on someone's head.
In this kind of situation where the occurrence of event is very rare Poisson distribution may be used.

Poisson distribution is the limiting case of binomial distribution. Under the following conditions binomial distribution tend to Poisson distribution.

1. There are infinite number of trials; $n \rightarrow \alpha$
2. Probability of getting an event is very small $p \rightarrow 0$
3. The product 'np' is constant $np = \lambda$

Poisson Distribution

If X is a random variable following a Poisson distribution, it is denoted as $X \sim \text{Poisson}(\lambda)$, where λ is the average number of occurrences in the interval.

The probability mass function (PMF) for a Poisson distribution is:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

Where:

- λ is the average rate of occurrence.
- e is the base of the natural logarithm (approximately 2.71828).
- k is the number of occurrences.

Properties of Poisson Distribution

1. Mean $= \lambda$
2. Variance $= \lambda$
3. In Poisson distribution Mean $=$ Variance
4. Standard deviation $=$ Sqrt(Variance)

Example

Suppose a bookstore receives an average of 3 customer inquiries per hour. The number of inquiries in a given hour follows a Poisson distribution with $\lambda = 3$.

Let's calculate the probability of receiving exactly 4 inquiries in an hour:

$$P(X = 4) = \frac{3^4 e^{-3}}{4!} = \frac{81 \times 0.0498}{24} = 0.168$$

So, the probability of receiving exactly 4 inquiries in an hour is approximately 0.168.

How to perform Probability distributions using Python?

Comparison



BINOMIAL: FIXED TRIALS, YES/NO



POISSON: RARE EVENTS OVER TIME



NORMAL: CONTINUOUS DATA, BELL CURVE

Multiple Choice Questions (MCQs)

What is a Random Variable?

a) A variable that takes deterministic values.
b) A variable that always takes integer values.
c) A constant that never changes.
d) A variable that assigns numerical values to the outcomes of a random process.

Multiple Choice Questions (MCQs)

What is a Random Variable?

a) A variable that takes deterministic values.
b) A variable that always takes integer values.
c) A constant that never changes.
d) A variable that assigns numerical values to the outcomes of a random process.

Multiple Choice Questions (MCQs)

Q1: Which distribution has mean = variance?

A) Normal

B) Binomial

C) Poisson

Q2: Which is used for yes/no trials?

A) Poisson

B) Binomial

Multiple Choice Questions (MCQs)

Q1: Which distribution has mean = variance?

A) Normal

B) Binomial

C) Poisson ✅

Q2: Which is used for yes/no trials?

A) Poisson

B) Binomial ✅

Session Summary

In this session, we have discussed about:

- Understand the concept of random variable and probability distributions.
- Understand Normal, Binomial, and Poisson Distributions.
- Learn the formulas, attempt quizzes and solve numerical problems.
- How to work with probability distributions using Python!

Reference Links

- Top 15 Probability Distribution Questions for Data Science Interviews
- Analyzing loan data with binomial and poisson distribution
- Notebook
- 10 Must-Know Probability Concepts for Every Aspiring Statistician and Data Scientist
- Top 20 Interview Questions on Descriptive and Inferential Statistics
- 10 Must-Know Statistics Interview Questions (With Real Examples & Formulas!)